

WebApi Handson Solution

2. WebApi Hands-on Solution

Objectives Addressed

- * Install Swagger (Swashbuckle.AspNetCore) and view Web API listing on the browser.
- * Use Postman tool to test Web API methods.
- * Use Route and ActionName attributes.

Solution Implementation

1. Swagger Installation and Setup

First, install the NuGet package:

```
```bash
dotnet add package Swashbuckle.AspNetCore
```
```

Add the following code to your `Startup.cs` or `Program.cs` to configure Swagger:

```
```csharp
// In ConfigureServices (or builder.Services)
builder.Services.AddSwaggerGen(c =>
{
 c.SwaggerDoc("v1", new Microsoft.OpenApi.Models.OpenApiInfo
 {
 Title = "Swagger Demo",
 Version = "v1",
 Description = "TBD",
 TermsOfService = new Uri("http://www.example.com"),
 Contact = new Microsoft.OpenApi.Models.OpenApiContact
 {
 Name = "John Doe",
 Email = "john@xyzmail.com",
 Url = new Uri("http://www.example.com")
 },
 },

```

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```
License = new Microsoft.OpenApi.Models.OpenApiLicense
{
 Name = "License Terms",
 Url = new Uri("http://www.example.com")
}
});
});

// In Configure (or app pipeline)
app.UseSwagger();
app.UseSwaggerUI(c =>
{
 c.SwaggerEndpoint("/swagger/v1/swagger.json", "Swagger Demo v1");
});
````
```

2. Using Postman & Route Attributes

Postman can be used to send HTTP requests to the application.

Below is the setup for an Employee Controller that modifies the Route attribute.

```
````csharp
using Microsoft.AspNetCore.Mvc;
using System.Collections.Generic;

namespace WebApi_Handson.Controllers
{
 // Modifying the route to 'Emp' as requested
 [Route("api/Emp")]
 [ApiController]
 public class EmployeeController : ControllerBase
 {
 // GET api/Emp
 [HttpGet]
 public ActionResult<IEnumerable<string>> GetEmployees()
 }
}
```

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```
{
 return new string[] { "Employee1", "Employee2" };
}

// GET api/Emp/ById/5 (Using ActionName equivalent via Route template)
[HttpGet("ById/{id}", Name = "GetEmployeeById")]
public ActionResult<string> GetEmployee(int id)
{
 return $"Employee {id}";
}
}
```

### ## Testing Output

1. **Swagger:** Navigating to `https://localhost:<port>/swagger` displays the UI with "Swagger Demo" title and API listings. Executing the GET method will return a successful 200 OK response with the JSON array.
2. **Postman:** Sending a GET request to `https://localhost:<port>/api/Emp` in Postman will list the employees in the 'Body' tab, and the 'Status' will read `200 OK`.